Gautam Solar Private Limited

IPQC Check Sheet

Document No.	GSPL/IPQC/IPC/003
Issue Date	01/12/2024
Rev. No./Rev.Date	01/30-08-2025

Date :- 01/09/2006 | Time :- 8:1 | Shift 2. | Po.no.:-

Sr.N	Stage	Check point	Quantum of Check Sample Size	Frequency	Acceptance Criteria	Monitoring Result	Remarks,if an
1	Shop Floor	Temperature	once	per shift	Temp. 25±3°C	Time 8:10 Pm 26.3°C	
		Humidity	once	per shift	RH ≤60%	Time 8:10 Pm 55°C	
2	Glass Loader	Glass dimension(L*	once	per shift	As Per PO	2376x1128x2.0mm	
		Appearance(Visual)	once	per shift	Glass Broken, Crack, Scratches	ok	
3	EVA/EPE Cutt	EVA/EPE Type	once	per shift	As per approved BOM	EP304	
		EVA/EPE dimension(	once	per shift	As per Specification	2378x1125x0.70mm	
		EVA/EPE Status	once	per shift	Not allowed dust & foreign part	Mfg: 26-12-2025	
4	Eva/EPE Sold	Soldering Tempratu	Once	per shift	As per specification and Should	415	
5	Cell Loading	Cell Manufacturer	once	per shift	Refer Process Card	25.70°C Solar Space	
		Cell Size(L*W)	once	per shift	Refer Process Card	105.09x182.41MM	
		Cell Condition	once	per shift	Free From dust,finger spot,color var	ok	
		Cleanliness of Cel	once	per shift	No unwanted or waste material s	clean	
		Verification of Pr	once	per shift	ATW Stringer Specificatio	ATW STRINGER - OK	
		Cell Cross cutting	once	per shift	Both side cutting should be equ	ok	
6	Tabber & str	Verification of Pr	once	Month	ATW Stringer specificatio	ATW Temp - All ok	
		Visual Check after	once	1 String/TS	TS Visual Criteria	TS01A: OK; TS01B: OK; TS02A: OK; TS02B: OK; TS03A: ok; TS03B: OK; TS04A: OK; TS04B: OK	
		EL Image of String	once	1 String/TS/	TS EL Criteria	TS01A: OK; TS01B: OK; TS02A: OK; TS02B: OK; TS03A: OK; TS03B: OK; TS04A: OK; TS04B: OK	
		String length	once	1 String/Str	Refer Process Card	TS01A: 1163; TS01B: 1162; TS02A: 1163; TS02B: 1164; TS03A: 1164; TS03B: 1163; TS04A: 1163; TS04B: 1163	
		Cell to Cell Gap				TS01A: 0.78; TS01B: 0.9; TS02A: 0.80; TS02B: 0.80; TS03A: 0.74; TS03B: 0.74; TS04A: 0.81; TS04B: 0.80	

		Verification of Sold	2 cell eac	per shift	Peel Strength ≥1N	Ribbon peel strenght OK					
7	Auto bussing	String to String Gap	once	per shift	Refer Process Card & Module Drawing	1.5 mm					
		Cell edge to Glass e	once	per shift		TOP: 1.5 MM					
						Bottom: 18.56 MM					
						Sides: 13.21 MM					
		Soldering Peel Stren	once	per shift	≥2N	Busbar peel test OK					
		Terminal busbar to e	once	per shift	132 Cell module drawing	132 cell Module					
					Refer Module Drawing: GSPL/	3.15 MM					
		Soldering Quality of	Every 4h	per shift	No Dry/Poor Soldering		OK OK OK				
		Top & Bottom Creepag	Every 4h	per shift	Creepage distance should be as per		T-12.35/B-11.24 I T-12.3				
		Verification of Proc	once	per shift	Specification for Auto Buss	OK					
		Quality of auto tapi	Every 4h	per shift	Taping should be proper,no Cell Shi		OK				
8	Auto RFID Lo	Position verificatio	Every 4h	per shift	Should not be tilt		OK OK OK OK				
9	EVA/EPE cutti	EVA/EPE Type	once	per shift	EVA	EP304					
		EVA/EPE dimension(L*	once	per shift	As per Specification	2378×1125×0.70mm					
		EVA/EPE Status	once	per shift	Not allowed dust & foreign particle	26-12-2025					
10	Back Glass Lo	Glass dimension(L*W*	once	Per shift	As per PO	2376×1128×2.0mm					
		No. of Holes/ Holes	once	Per shift	3 hole with dimension 12mm±0.5mm	3 holes 12.71mm 12.78mm 12.81mm					
11	Auto Busbar F	Visual Inspection	5 pieces	per shift	No cracks/ breaks in busbar &prope	OK OK OK OK O					

12	Pre lamina	EL Inspection a	5 piece	per shift	Pre EL Inspection Cr	S.No	GS04890TG3562555674	
						S.No	GS04890TU73562555678	
						S.No	GS04890TG3562555680	
					Pre EL Visual Criter	S.No	GS04890TG73562555682	
						S.No	GS04890TG3562555672	
13	String Rew	cleaning of rew	once	per shift	Rework Station should be C	clean and wet		
		Soldering Iron	once	per shift	400±30°C	Time: 8:10 Pm 418		
14	Module Rew	Method of Rewor	once	per shift	As per WI (GSPL/P/WI/012)	manual		
		Cleaning of Rew	once	per shift	Rework Station should be C	clean and wet		
		Soldering Iron	once	per shift	400±30°C	Time: 8:10 Pm 426		
15	Laminator	Monitoring of L	once	per shift	Process Parameter of	OK		
		Cleaning of Dia	once	24h	Diaphragm/Release sheet sh	claen		
		Peel of Test b/	All pos	All lamina	E/G ≥60N/cm E/B≥60N/cm	Refer Document GSPL/IPQC/QC/001		
		Gel Content Tes			75to 95%	Refer Document GSPL/IPQC/QC/001		

16	Auto Tap	Visual Check	5 piec	per shif	Check Tape Removing S	OK OK OK O					
17	Auto Edg	Trimming Qua	5 piec	per shif	Excess layer from the	S.No.	GS04890155044264874				
						S.No.	GS04890KG104426449 43				
						S.No.	GS04890120044264866				
						S.No.	GS04890KG04426448 69				
						S.No.	GS04890120044264183				
		Trimming Bla	once	per mont	Worn out not allowed	OK					
18	90° Visu	Visual Inspe	5 piec	per shif	Post Lam Visual I	S.No	GS04890KG04426461 10				
						S.No	GS04890120044264612 0				
						S.No	GS04890KG04426461 61				
						S.No	GS04890KG04426461 81				
						S.No	GS04890100442646183				
19	Framing	Glue uniform	1 set	per shif	Should be uniform,Bac	OK					
		Short Side G	once	Per shif	Fill as per Specifica	22.113gM					
		Long Side Gl	once	Per shif		26.113gM					
		Anodizing Thi	once	Per shif	≥15 micron	215 Micron					

20	Junction Bo	Junction Box(Conn	once	Per shift	As per Process Card & module	300 mm						
		Silicon Glue Weig	once	Per shift	21±6 gm	26.193g						
21	Auto JB Sol	Max Welding time	once	Per shift	As per Specification	2 sec.						
		Soldering current	once	per shift	As per Specification	17A						
		Soldering Quality	once	per shift	Welding area should be fully	Good						
22	JB Potting	A/B Glue Ratio	once	Per shift	As per Specification	Refer Document GSPL/IPQC/QC/011						
		Potting material	once	Per shift	21±6 gm	1:1						
		Nozzle Changing	once	every 6h	Should be changed after 6 hou	8:10 AM			2:10 PM			
23	OLE Potting	Visual Check	once	5 piece	Potting should be properly fi	OK O.K O.K						
24	Curing	Temperature	once	per shift	25±3■	25.4°C						
		Humidity	once	per shift	≥50%	67°C						
		Curing Time(h)	once	Per shift	≥4 hours	4 hrs						
25	Buffing	Corner Edge,Buffi	5 pieces	per shift	Should not be sharp & No worn	OK OK OK OK						
26	Cleaning	Module should be	5 pieces	per shift	Post Lam Visual Cr	S.No	GS04890120044264 6120					
						S.No	GS04890KG044264 6161					
						S.No	GS04890KG044264 6181					
						S.No	GS04890100442646183					
						S.No	GS04890190442646192					

27	Flash Test	Ambient Temp.	once	per shift	25±3■	26.3°c				
		Module Temp.	once	per shift	25±3■	24.65°C				
		Sunsimulator Cal	once	12h	Sunsimulator must be calibr	GS04890KG1044268001 9				
		Validation	once	every 6h	As per GSPL/QA/S/11	O.K OK				
		Silver Reference	once	Two weeks	Should be same as orignal E	OK				
28	Hipot Test	DCW/IR/Ground C	5 pieces	per shift	≤50µA , >40MΩ.m2 ,(0-100) m	GS04890KG044264908	1.3 A	2.849	2.015m	
						GS0489012070442610 02	1.3 A	2.828M	2.277m	
						GS04890TDI16252471	1.1 A	3.040M	4.111m	
						GS04890KG044264616 9	1.2 A	3.002M	2.013m	
						GS04890K01044264621 0	1.1 A	3.203M	4.428m	
29	Post EL Te	Voltage & Curren	once	Shift	As per WI (GSPL/P/WI/027)	50.3V, 7.5A				
		EL Inspection an	5 pieces	per shift	Post EL Inspection Cr	S.No	GS04890160442646216			
						S.No	GS04890KG044267622 4			
						S.No	GS04890K00442656165			
					Post EL Visual Criter	S.No	GS04890129044265616 8			
						S.No	GS04890KG044265612			
30	RFID	RFID Position	once	per shift	As per Process card	centre left side				
		Cell & Module Ma	once	per shift	As per BOM and Process card	Feb. 2026				

31	Final Visua	Visual Inspection	5 pieces	per shift	Post lam visual inspect	S.No.	GS04890K0044265616 5	
						S.No.	GS0489012904426561 68	
						S.No.	GS04890KG044265612	
						S.No.	GS04890KG044265617 2	
						S.No.	GS04890KG044265612 5	
		Backlabel	5 pieces	per shift	Air bubble,Tilt,Misprint,folde	S.No	GS0489019044265617 7	
						S.No	GS04890KM044265618	
						S.No	GS04890KG044265618 6	
						S.No	GS0489010044265616 2	
						S.No	GSW4890LG044265617 8	
32	Dimension m	L*W and Module Pr	once	per shift	As per Module drawing (±1mm)	(2382 x 1134 x 30)mm		
		Mounting Hole X &	once	Per shift		(1400 x 1091)mm		
		Diagonal Differen	once	Per shift	≤3mm	0.01 mm		
		Corner Gap	once	Per shift	As per visual Inspection crite	0.02 MM		
		JB Cable length	once	Per shift	As per Process card	300 mm		
33	Packaging	Packaging Label	once	Per shift	WI For Packaging	Ok		
		Content in Box	once	Per shift		OK		
		Box Condition	once	Per shift	Damage,dull printing,wet boxes	OK		
		Wooden Pallet dim	once	Per shift	Should not be less than module	(2400 x 1050 x 146) MM		
Checked By						Reviewed By		